AI 2002
Artificial Intelligence
Lecture 12

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FAST NUCES CFD

Knowledge

Humans know things …

The knowledge helps them to do various tasks.

- **The knowledge has been achieved**
  - not by purely reflex mechanisms
  - but by the processes of **reasoning**
- In AI, the example is **knowladge-based agent** which contains **set of sentences** referred as **knowledge-base.**

Knowledge Based Agents

- <u>**For a generic knowledge-based agent:**</u>
- A **percept is given** to the agent.

- The agent **adds the percept** to its knowledge base.

- **Perform best action** according to the knowledge base.

- Tells the knowledge base that it has in fact **taken that action**.

Knowledge Based Agents

function KB-AGENT(*percept*) **returns** an *action*
  persistent: KB, a knowledge base
    t, a counter, initially 0, indicating time

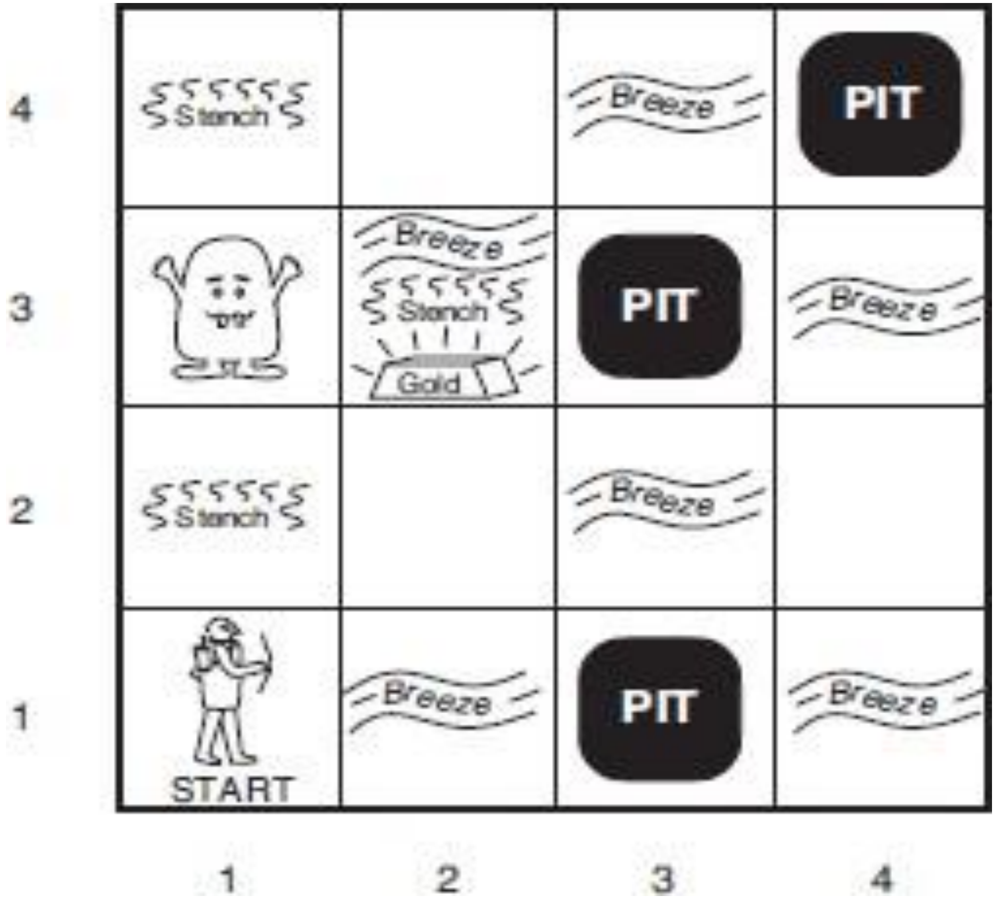
  TELL(KB, MAKE-PERCEPT-SENTENCE(*percept*, t))
  action ← ASK(KB, MAKE-ACTION-QUERY(t))
  TELL(KB, MAKE-ACTION-SENTENCE(*action*, t))
  $t \leftarrow t + 1$
  return *action*

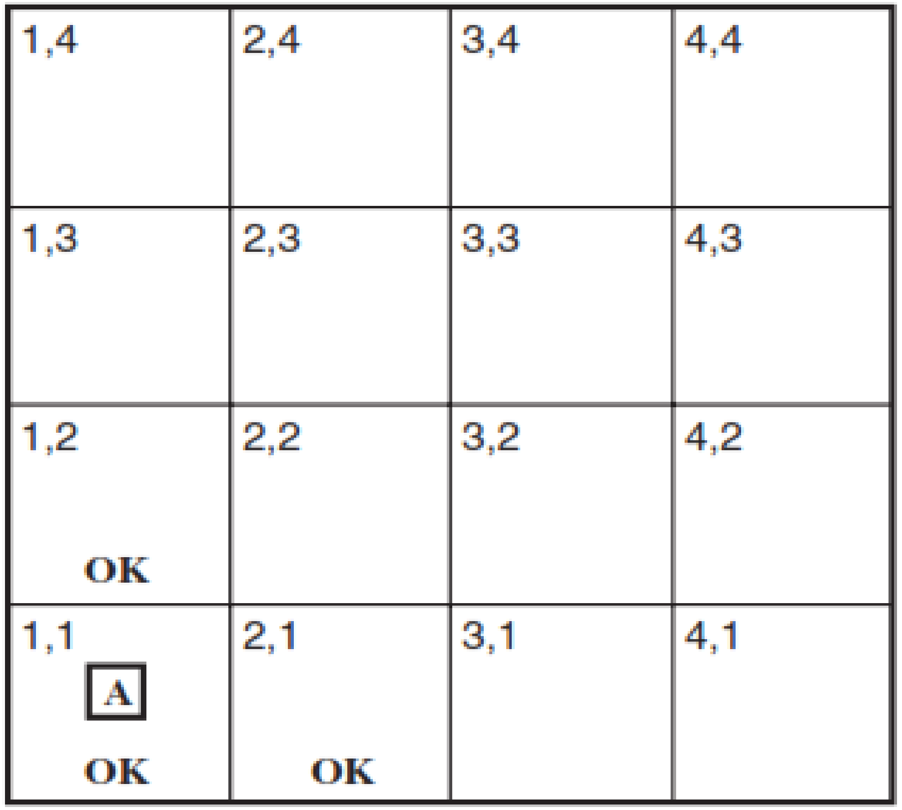
constructs a **sentence** asserting that the agent ***perceived the given percept*** at time ***t***

constructs a sentence that asks ***what action should be done*** at time ***t***

constructs a sentence that ***the chosen action was executed*** at time ***t***

The Wumpus World Problem





The Wumpus World Problem

The PEAS description for Wumpus World:

Performance measure:

- +1000 for climbing out of the cave with the gold,
- –1000 for falling into a pit or being eaten by the Wumpus,
- –1 for each action taken
- –10 for using up the arrow
- **Environment:**
- A 4×4 grid of rooms. The agent starts in the square labelled [1,1], facing to the right.

The game ends either when the agent dies or when the agent climbs out of the cave.

The Wumpus World

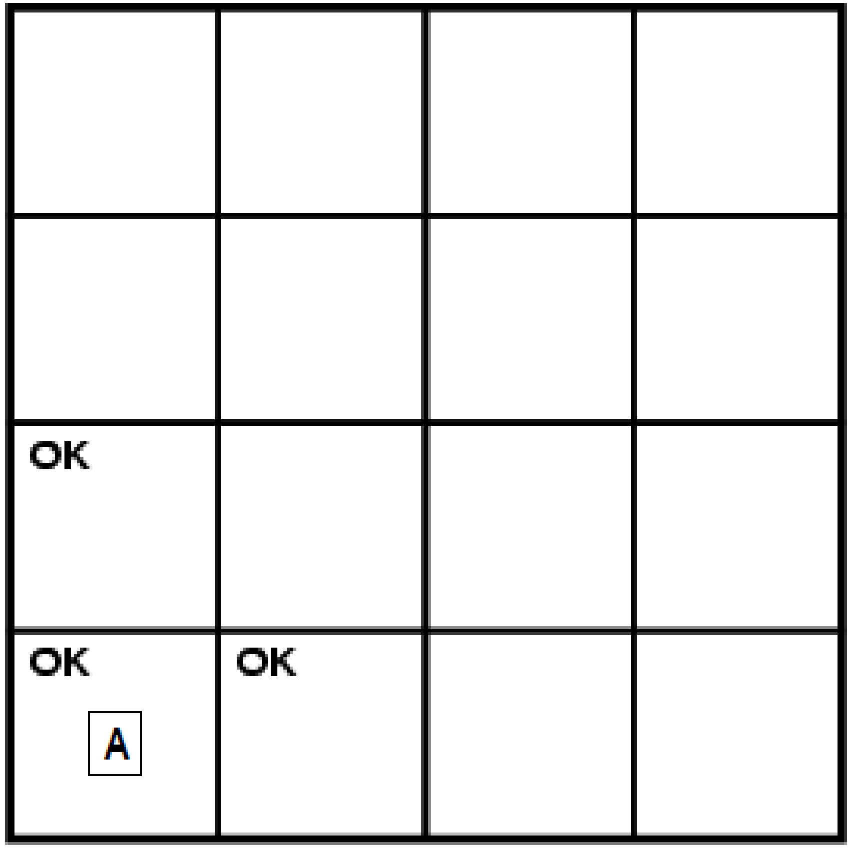
- **The PEAS description for Wumpus World:**

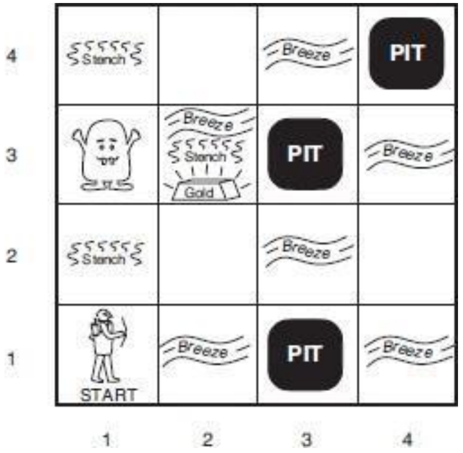
Actuators:

  - The agent can move *Forward*, *TurnLeft by 90◦*, *TurnRight by 90◦*, grab, shoot

- **Sensors:**
  - The square adjacent directly (not diagonally) to the square containing **Wumpus**, the agent will perceive a ***Stench***.
  - The squares adjacent to a **pit**, the agent will perceive a ***Breeze***.
  - The square with **gold**, the agent will perceive a ***Glitter***.
  - An agent **walks into a wall**, it will perceive a ***Bump***.
  - When the **Wumpus is killed**, it emits a woeful ***Scream***.

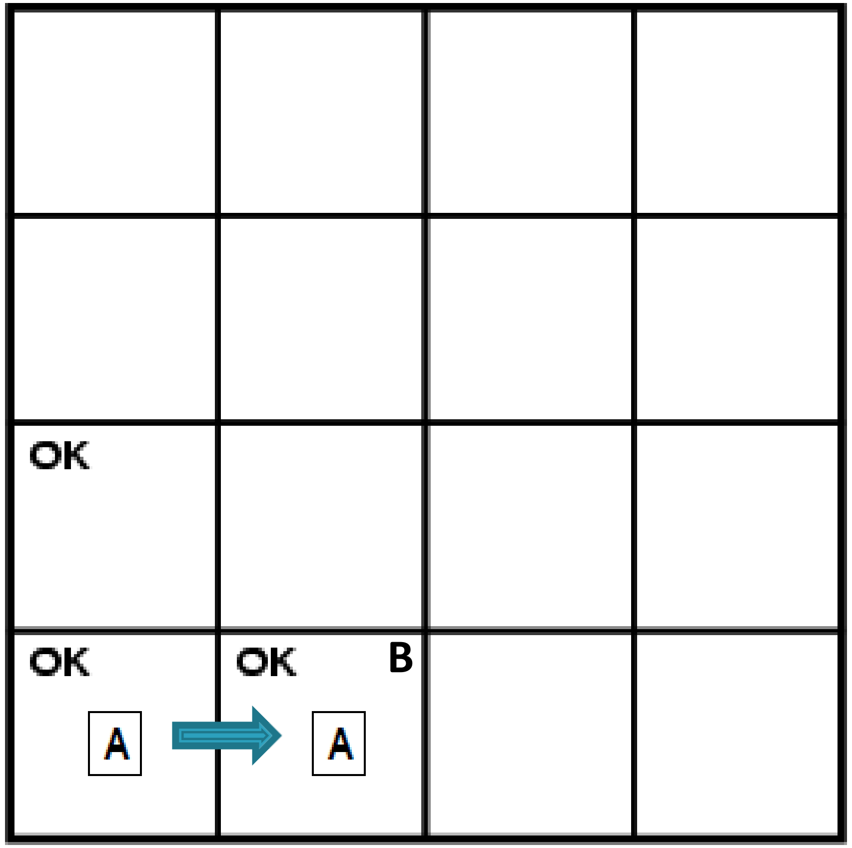
The Wumpus World

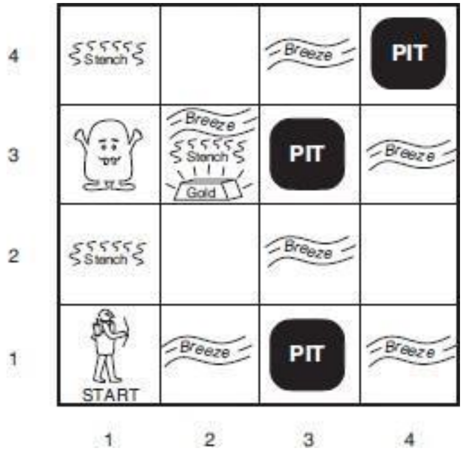




A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

The Wumpus World





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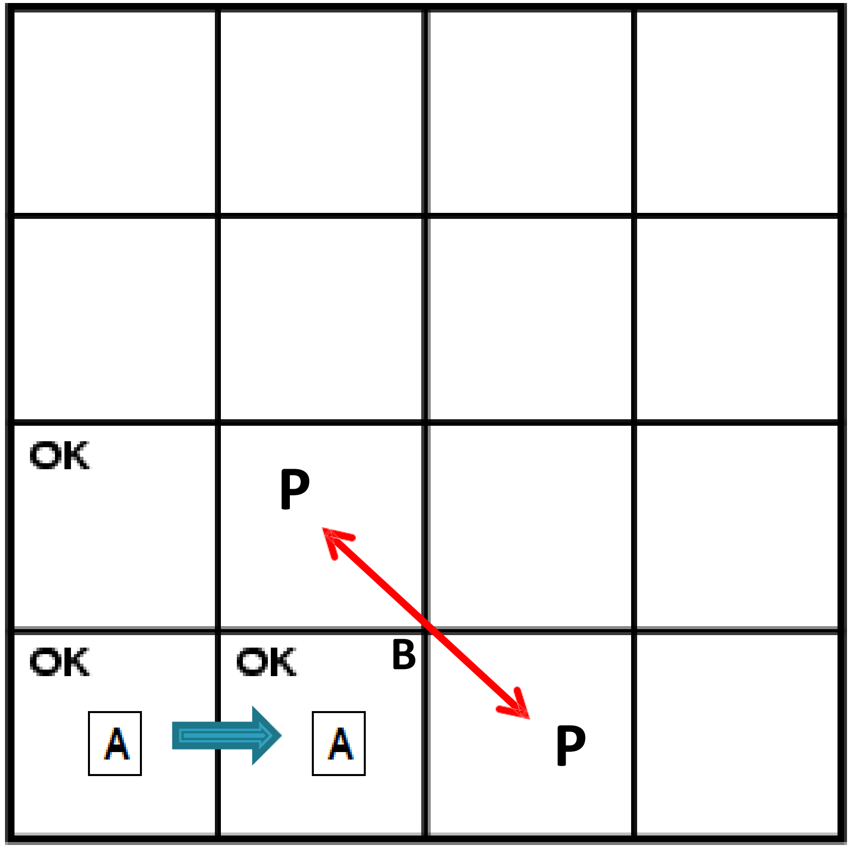

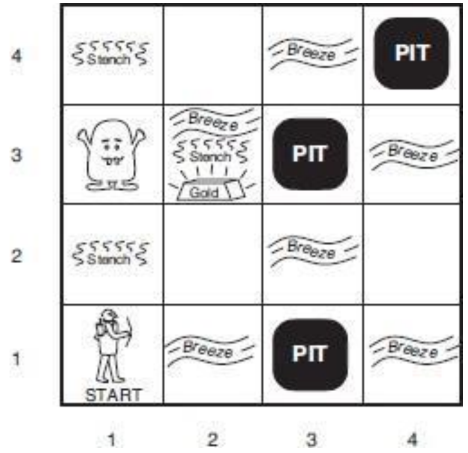

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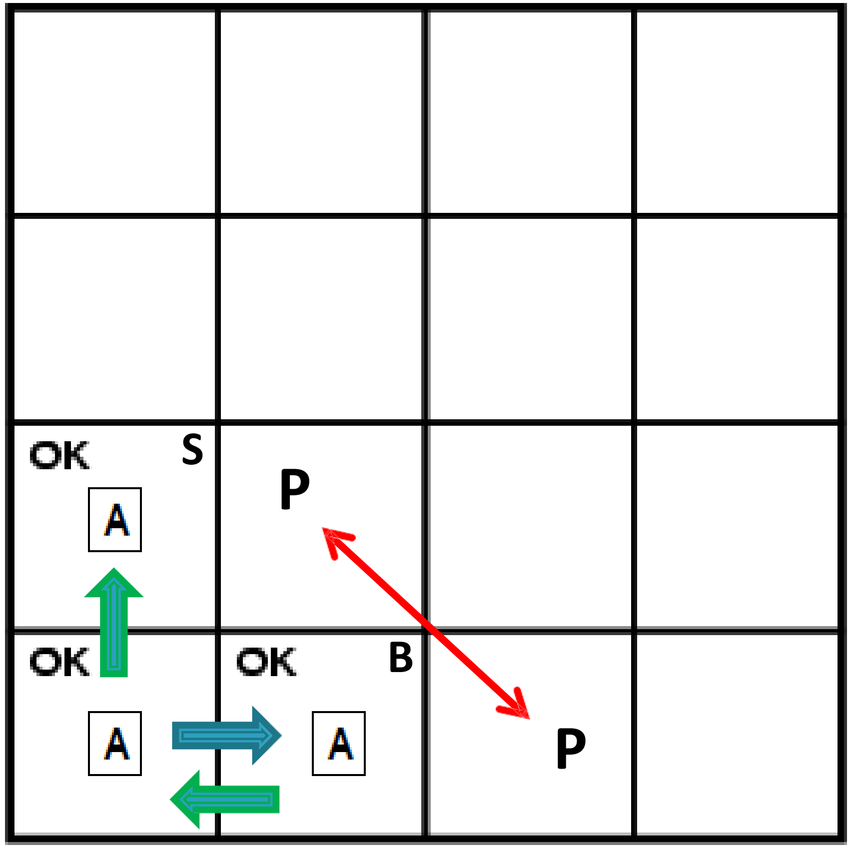

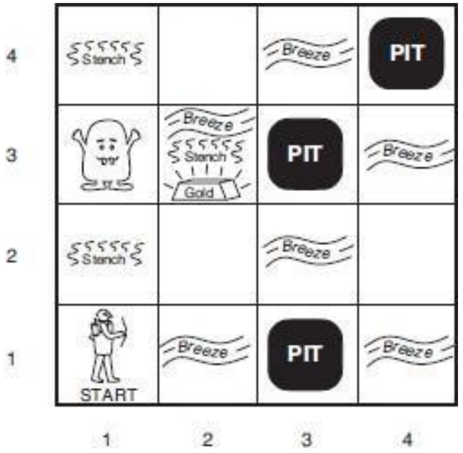

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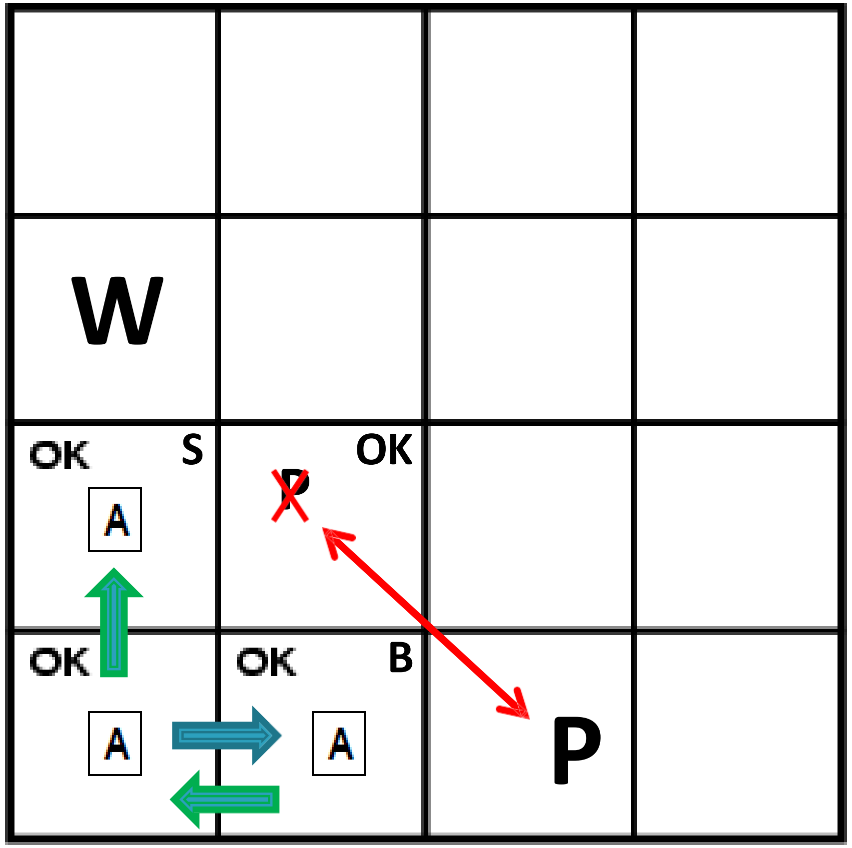

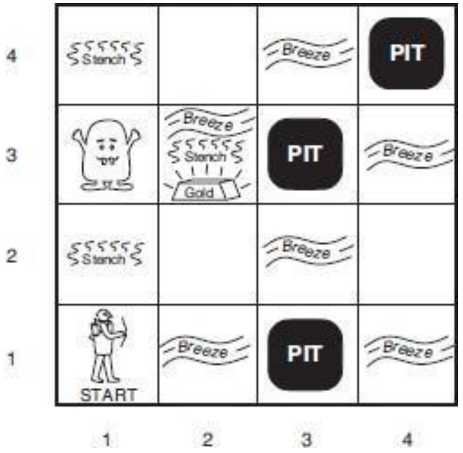

- **A** = *Agent*
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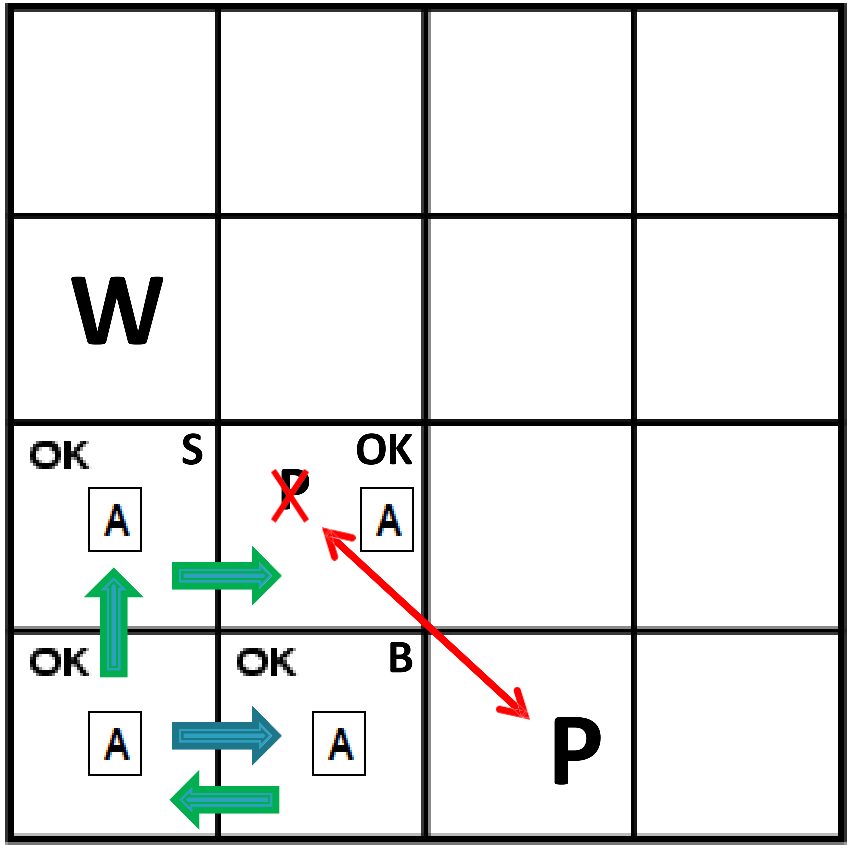

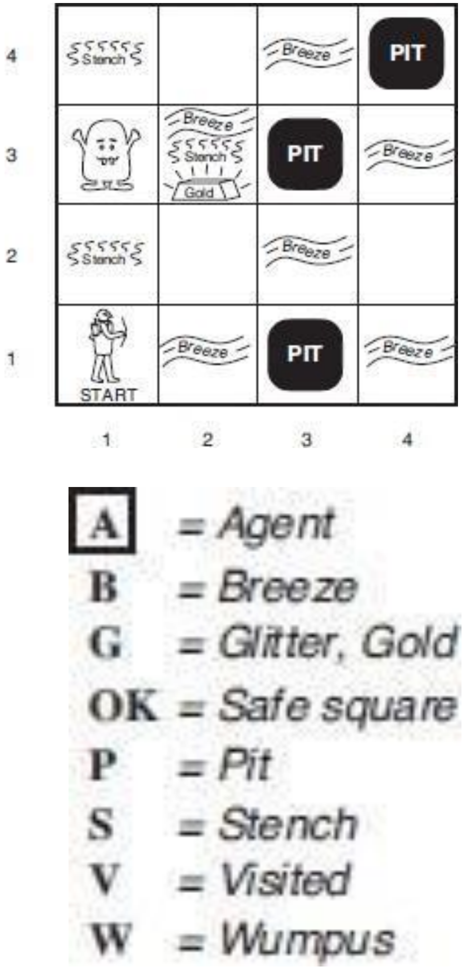

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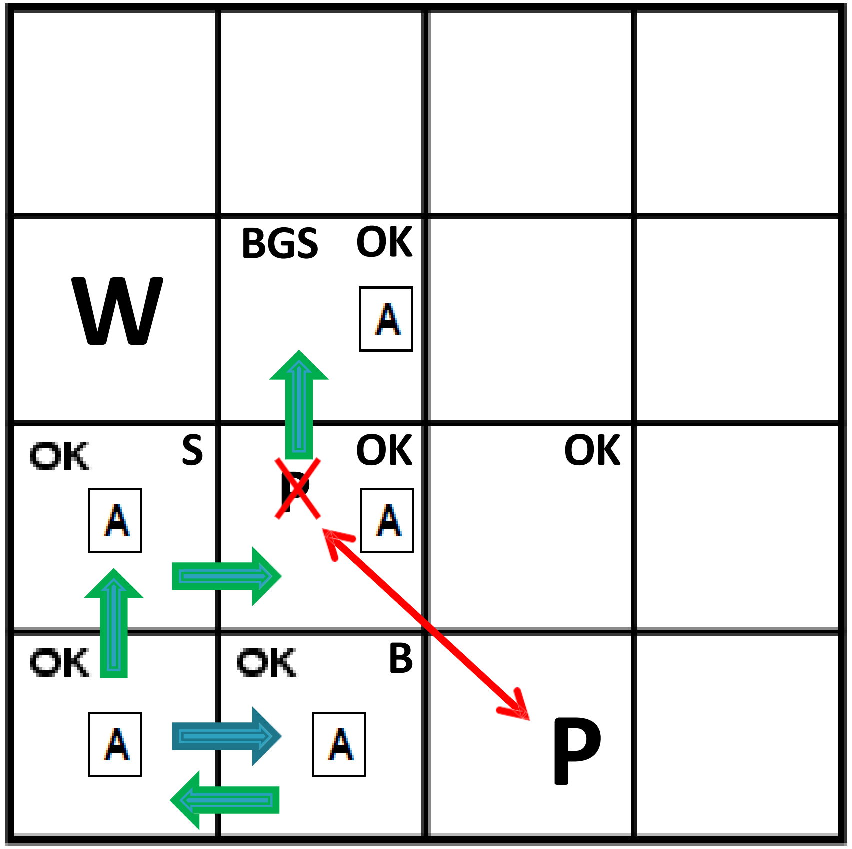

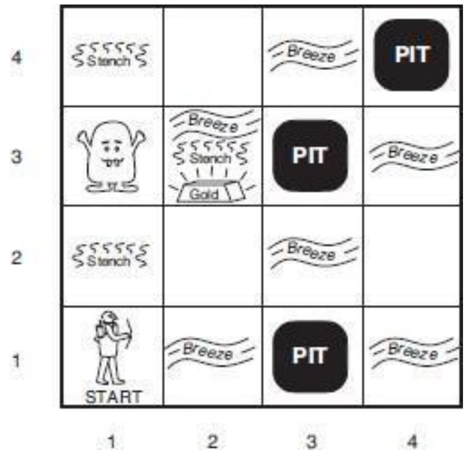

A = *Agent*
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Logic

- How to represent these sentences?
- The knowledge bases consist of sentences.
- **Logic**, a formal language, is the solution --- a way of manipulating expressions in the language.
- Logic has
  - Syntax
  - Semantics

Logic

- **Syntax:**
  - *What expressions are legal* --- what are allowed to write down.
- The notion of syntax is clear enough with the example: "$x + y = 4$" is a well-formed sentence, whereas "$x4y+ =$" is not.
- **Semantics:**
  - *What legal expression means* --- meaning of sentences the sentence "$x + y = 4$" is **true** in a **world** where x is 2 and y is 2, but **false** in a **world** where x is 1 and y is 1.
- *Syntax is a form and semantics is the content.*

Logic

- **Semantics:**
- The semantics defines the **truth** of each sentence with respect to each **possible world.**
- The term **model** can be used in place of **“possible world.”**
- If a sentence $\boldsymbol{\alpha}$ is true in model $\boldsymbol{m}$, we say that $\boldsymbol{m}$ **satisfies** $\boldsymbol{\alpha}$ or sometimes $\boldsymbol{m}$ is a **model** of $\boldsymbol{\alpha}$.
- The notation $\boldsymbol{M(\alpha)}$ --- the set of all **models** of $\boldsymbol{\alpha}$.

Entailment

- means that **one thing follows from another**:

$$\alpha \models \beta$$

- *if and only if, in every model in which* ***α*** *is true,* ***β*** *is also true*. We can write

$$\alpha \models \beta \text{ if and only if } M(\alpha) \subseteq M(\beta)$$

- The notation ⊆ means that: if $\alpha \models \beta$, then ***α*** is a *stronger assertion than* ***β***
- Example:

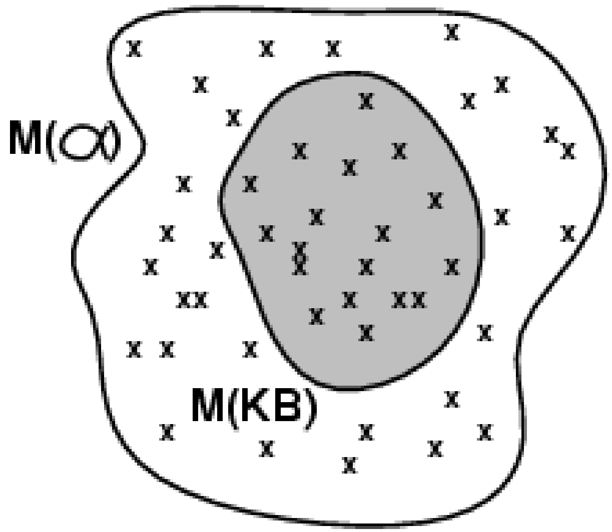
α = The dog is sleeping on the couch

β = The dog is sleeping.

Entailment Example

- We say $\boldsymbol{m}$ is a model of sentence $\boldsymbol{\alpha}$ if $\boldsymbol{\alpha}$ is true in $\boldsymbol{m}$
- $\boldsymbol{M(\alpha)}$ is the set of all models of $\boldsymbol{\alpha}$

- Then $\boldsymbol{KB} \models \boldsymbol{\alpha}$ iff $\boldsymbol{M(KB)} \subseteq \boldsymbol{M(\alpha)}$

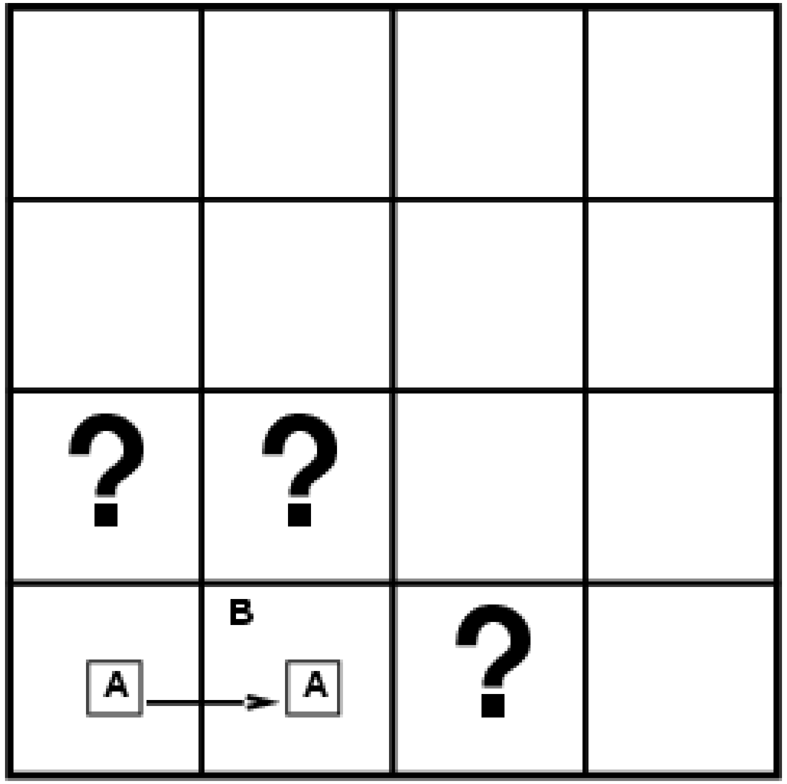


- **Example:**
- The sentence $\boldsymbol{x = 0}$ entails the sentence $\boldsymbol{xy = 0}$
  - In any model where $\boldsymbol{x}$ is zero, it is the case that $\boldsymbol{xy}$ is zero
- (regardless of the value of $\boldsymbol{y}$)

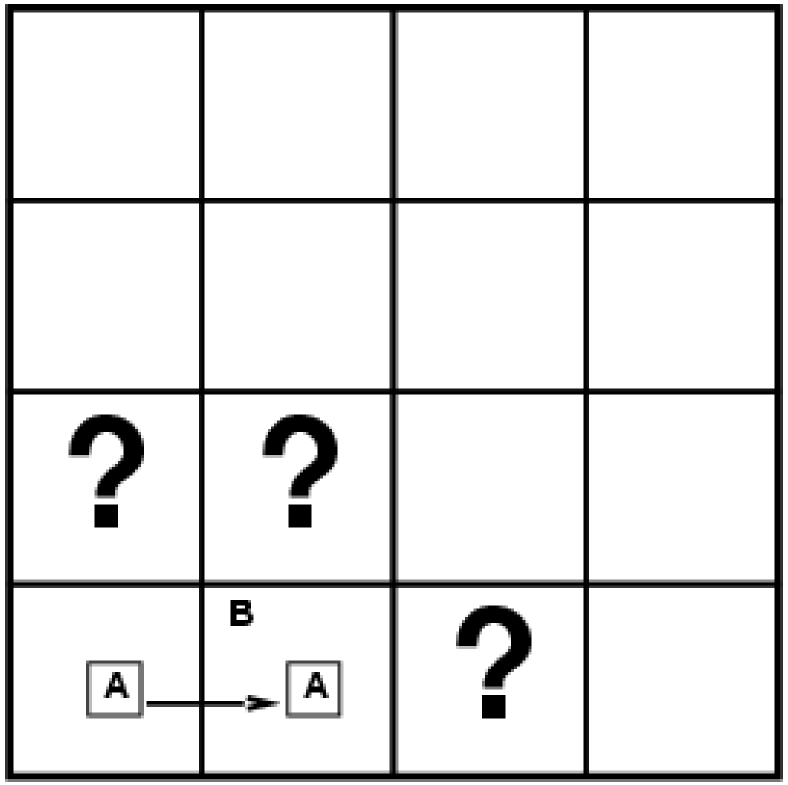
Entailment Wumpus World

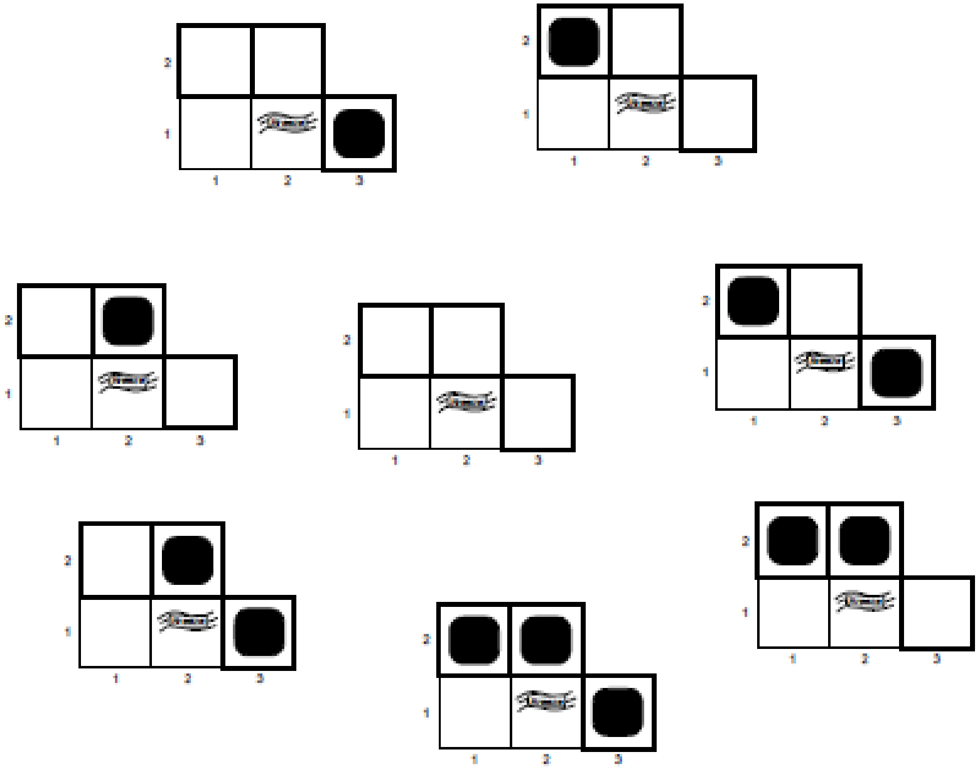
Situation after detecting nothing in [1,1], moving right, **breeze** in [1,2]

Consider possible models for **KB** assuming only pits



Entailment Wumpus World

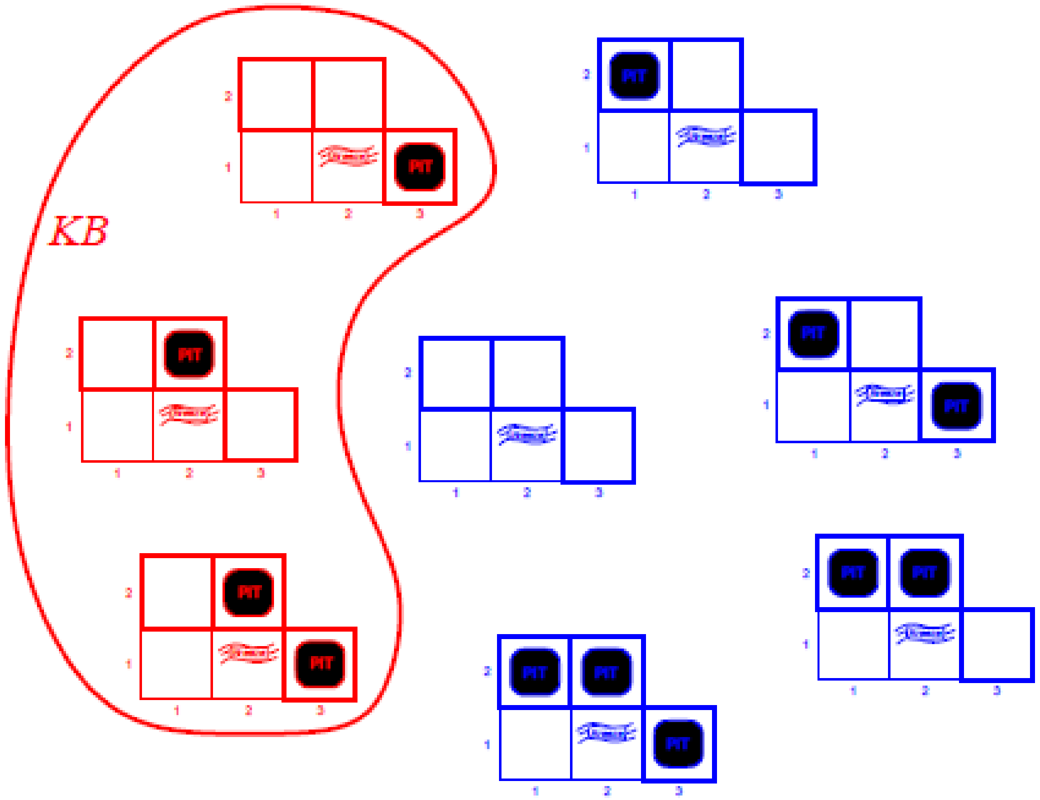




3 Boolean choices $\Rightarrow$ 8 possible models

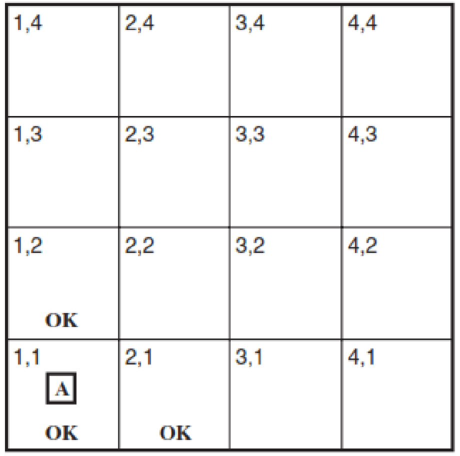
regardless of wumpus-world rules

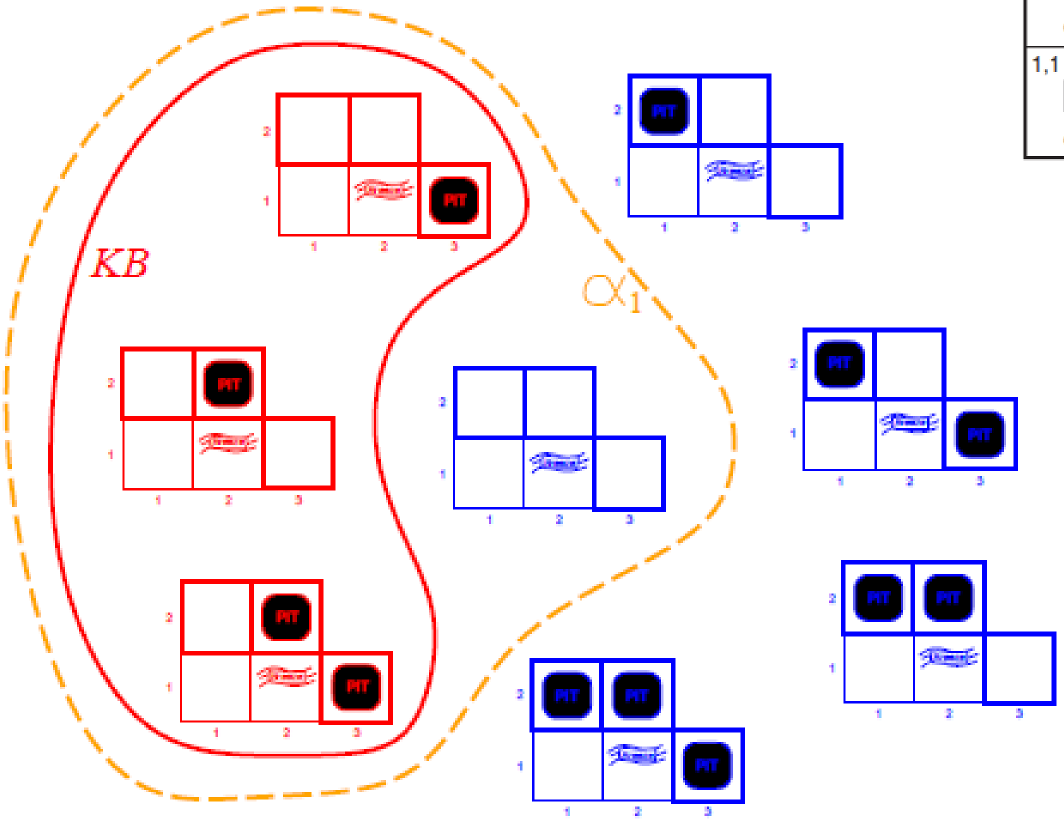
Entailment Wumpus World



KB = wumpus-world rules + observations

Entailment Wumpus World



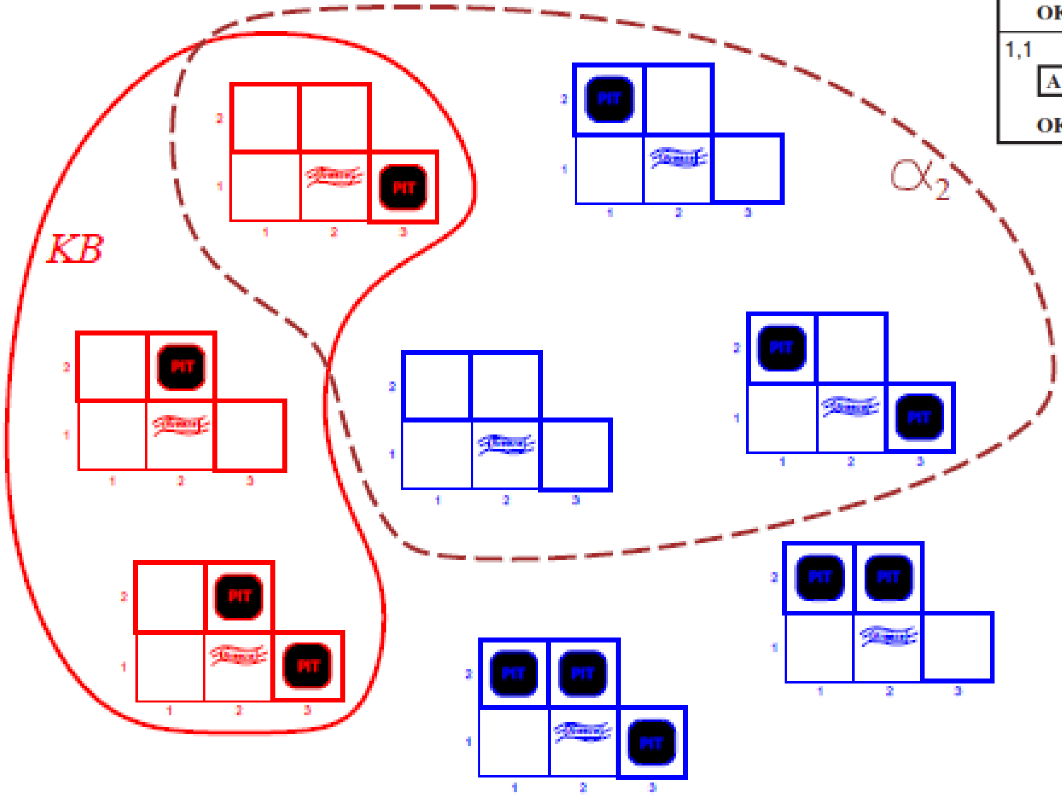


KB = wumpus-world rules + observations

α_1 = "[1,2] is safe", $KB \models \alpha_1$, proved by model checking

Entailment Wumpus World

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2	3,2	4,2
1,1 $\boxed{A}$ OK	2,1 OK	3,1	4,1



KB = wumpus-world rules + observations

α_2 = "[2,2] is safe", $KB \nvDash \alpha_2$

Inferences

- If an inference algorithm $\boldsymbol{i}$ can derive $\boldsymbol{\alpha}$ from $\boldsymbol{KB}$, we write

$$KB \vdash_i \alpha$$

- which is pronounced **“$\boldsymbol{\alpha}$ is derived from $\boldsymbol{KB}$ by $\boldsymbol{i}$”** or **“$\boldsymbol{i}$ derives $\boldsymbol{\alpha}$ from $\boldsymbol{KB}$.”**
- **Soundness:**
  - An inference algorithm that **derives only entailed sentences** is called **sound or truth preserving.**
  - Soundness is a highly desirable property.
- **Completeness:**
  - An inference algorithm is complete if it can derive any sentence that is **entailed**.

Logic

- We Discuss two types of logic
  1. Propositional Logic
     Which is relatively simple

  2. First Order Logic
     Which is more Complicated

Propositional Logic

- The **syntax** of propositional logic defines the *allowable sentences*.

What are the sentences?

- Sentence are well formed formulas
- **True** and **False** are sentences
- Propositional variables are sentences. P, Q, R, S etc.

Propositional Logic : Syntax

- The **atomic sentences** consist of a **single proposition symbol**.
- Each such symbol stands for a proposition that can be **True or Fal**se.
- The **complex sentences** are constructed from simpler sentences, using parentheses and **logical connectives**.
- There are five connectives in common use:
  - ¬ (not)
  - ∧ (and)
  - ∨ (or)
  - ⇒ (implies)
  - ⇔ (if and only if)

Propositional Logic:

If S is a sentence, $\neg S$ is a sentence (negation)

If S_1 and S_2 are sentences, $S_1 \wedge S_2$ is a sentence (conjunction)

If S_1 and S_2 are sentences, $S_1 \vee S_2$ is a sentence (disjunction)

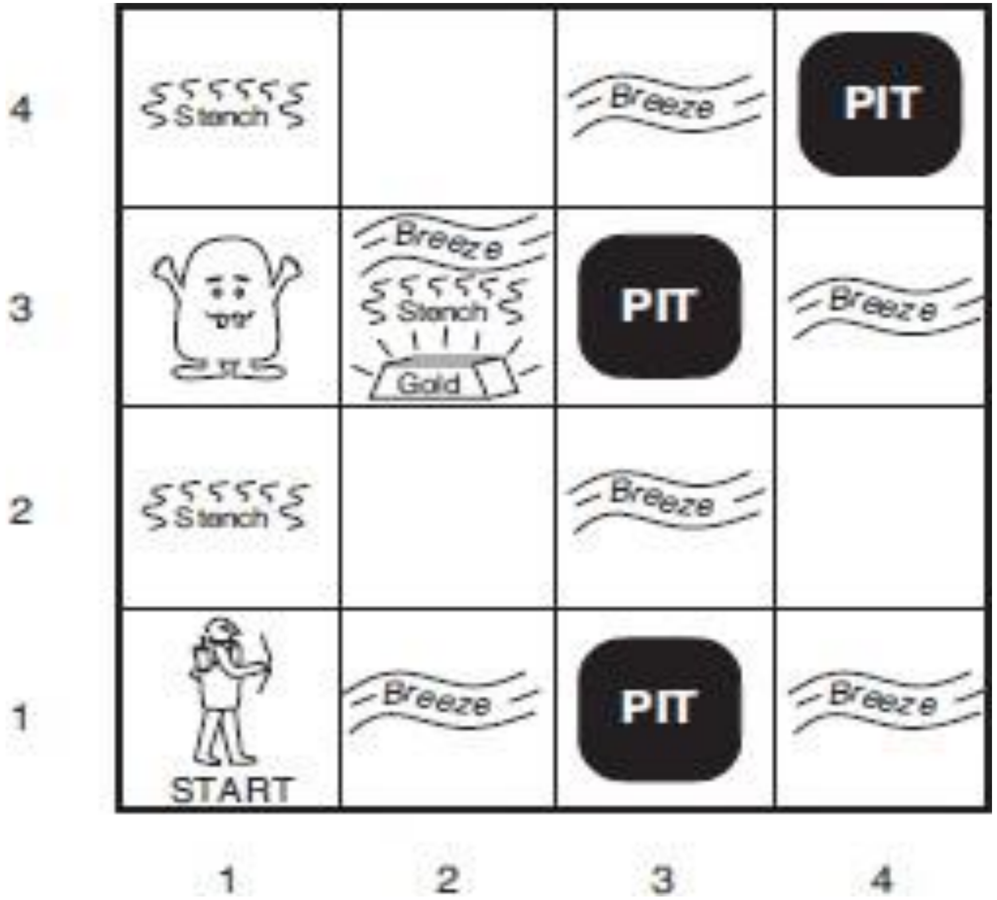
If S_1 and S_2 are sentences, $S_1 \Rightarrow S_2$ is a sentence (implication)

If S_1 and S_2 are sentences, $S_1 \Leftrightarrow S_2$ is a sentence (biconditional)

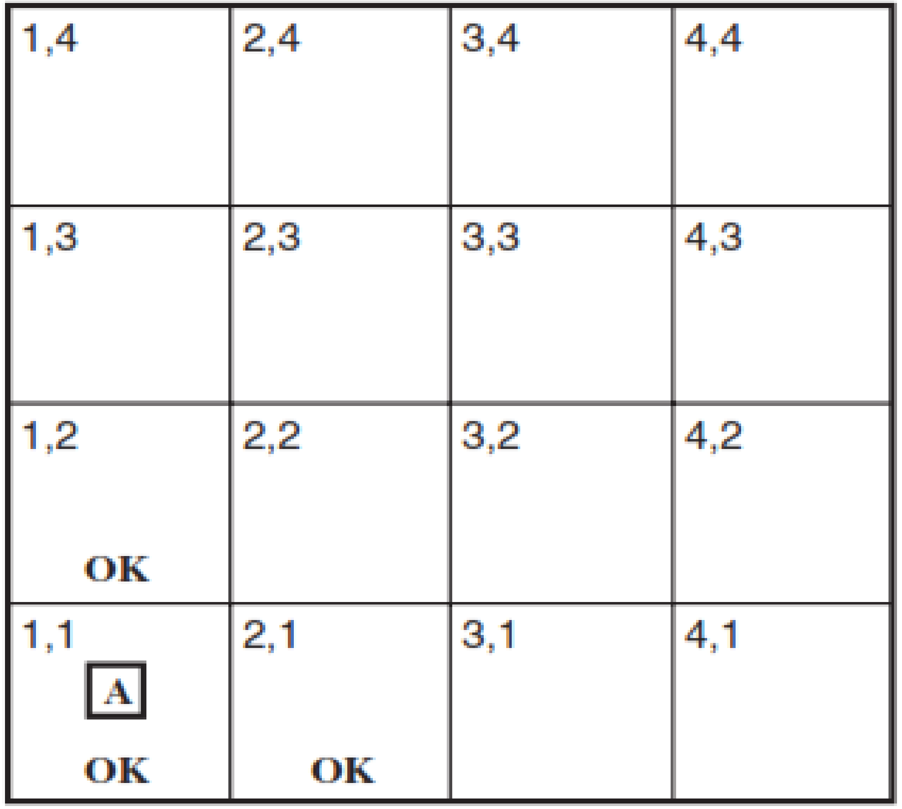
BNF

$$
\begin{array}{rcl}
Sentence & \rightarrow & AtomicSentence \mid ComplexSentence \\
AtomicSentence & \rightarrow & True \mid False \mid P \mid Q \mid R \mid \ldots \\
ComplexSentence & \rightarrow & (\ Sentence\ ) \mid [\ Sentence\ ] \\
& \mid & \neg\ Sentence \\
& \mid & Sentence \land Sentence \\
& \mid & Sentence \lor Sentence \\
& \mid & Sentence \ \Rightarrow\ Sentence \\
& \mid & Sentence \ \Leftrightarrow\ Sentence \\
\\
\text{OPERATOR PRECEDENCE} & : & \neg, \land, \lor, \Rightarrow, \Leftrightarrow
\end{array}
$$

Example



4
3
2
1
Stench
Breeze
PIT
Breeze
Stench
Gold
PIT
Breeze
Stench
Breeze
START
Breeze
PIT
Breeze
1
2
3
4

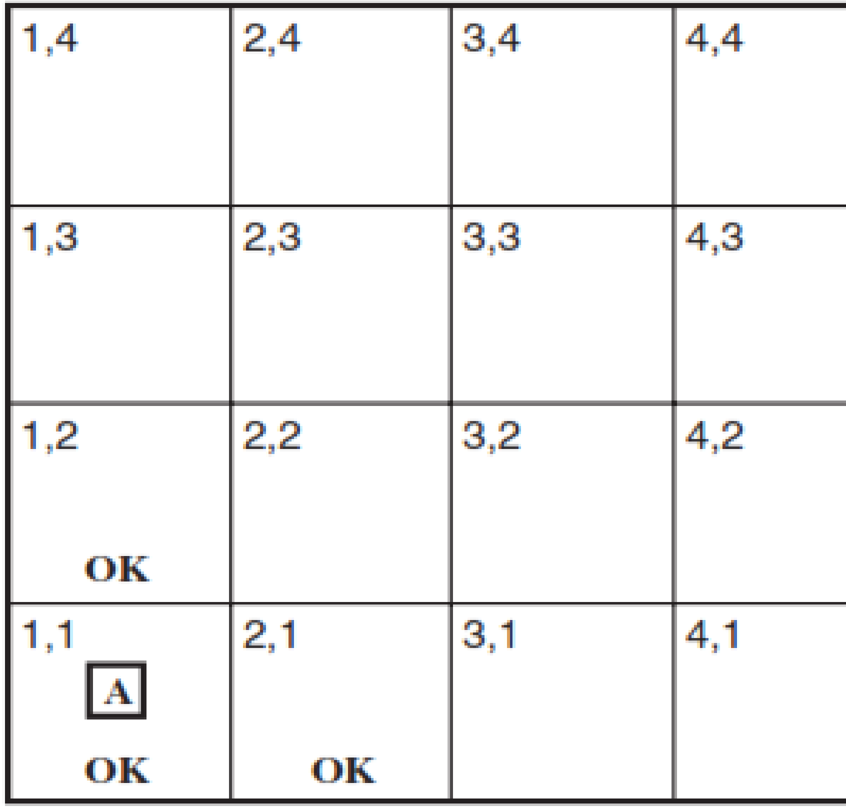


1,4
2,4
3,4
4,4
1,3
2,3
3,3
4,3
1,2
OK
2,2
3,2
4,2
1,1
A
OK
2,1
OK
3,1
4,1

Example:

- If there is an stench in the Square $S_{1,2}$, then there is the Wumpus in the Adjacent square.
- Solution:
  - Identify the syntax structure in the above statement
    - If
    - Or
- Propositional form:

$$S_{1,2} \rightarrow W_{2,2} \vee W_{1,3}$$

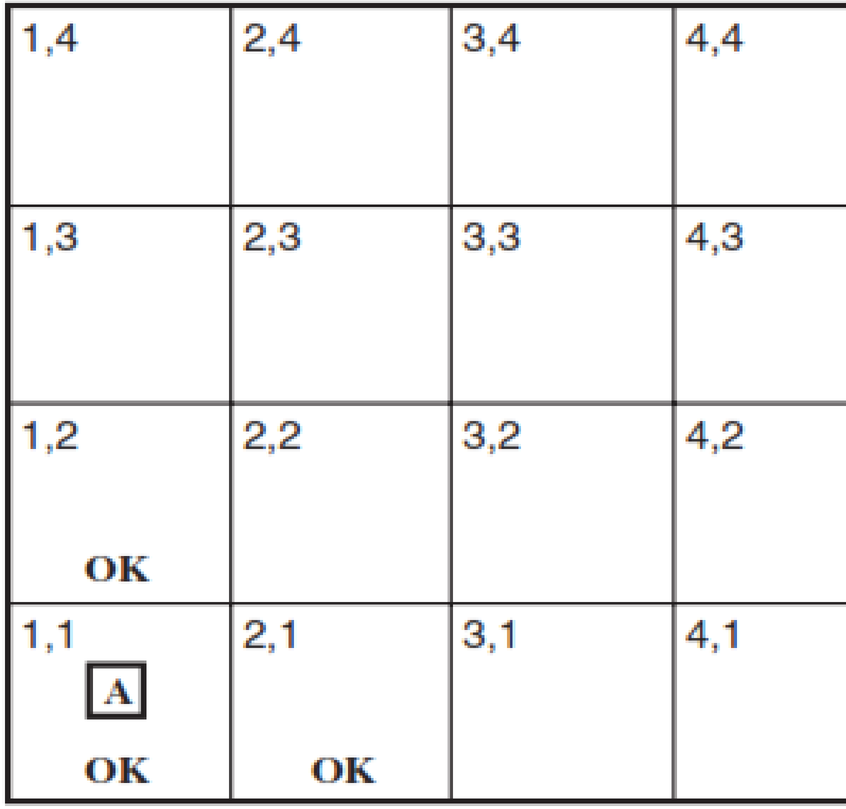


Example:

If there is an gold in the Square $S_{2,3}$, then there is the breeze and stench in the square $S_{3,3}$

- Solution:
  - Identify the syntax structure in the above statement
    - If
    - and
- Propositional form:

[redacted],2 $\to W_{2,2}$ [redacted],3



Propositional Logic Semantics

- **The semantics** defines **the rules** for determining the truth value of a sentence with respect to a particular model.
- In propositional logic, a model simply fixes the truth value—**true** or **false**—for every proposition symbol
- **For example**:
  - If the sentences in the knowledge base make use of the proposition symbols $\boldsymbol{P}_{1,2}$, $\boldsymbol{P}_{2,2}$, and $\boldsymbol{P}_{3,1}$, then one possible model is:

$$m_1 = \{P_{1,2} = false,\ P_{2,2} = false,\ P_{3,1} = true\}$$

Propositional Logic Semantics

- The semantics for propositional logic must specify **how to compute the truth value** of *any sentence, given a model.*
- **For Atomic sentences:**
- **True** is true in every model and **False** is false in every model.

The truth value of every other proposition symbol must be specified directly in the model.

  - For example, in the model $\boldsymbol{m_1}$ given earlier, $\boldsymbol{P}_{1,2}$ is false.

$$m_1 = \{P_{1,2} = false,\ P_{2,2} = false,\ P_{3,1} = true\}$$

Propositional Logic Semantics

- $\neg P$ is true iff P is false in m.
- $P \wedge Q$ is true iff both P and Q are true in m.
- $P \vee Q$ is true iff either P or Q is true in m.
- $P \Rightarrow Q$ is false unless P is true and Q is false in m.
- $P \Leftrightarrow Q$ is true iff P and Q are both true or both false in m.